

Retail Business Intelligence

Executive Decision Report & Capstone Presentation

RabTech Academy — Data Analytics & Business Intelligence Internship

Prepared from Tasks 02–05: KPI & Data Quality → Cleaning → EDA → Interactive Dashboard

Prepared by Vaibhav Gupta

2. Executive Summary

- The portfolio analyzes 12,575 cleaned retail transactions generating ■1,636,956 revenue.
- Average Order Value is ■130.18, with 25 unique customers and 69,828 units sold.
- Butchers is the highest-revenue category at ■217,020; Electric Household Essentials follows at ■214,638.
- 2024 is the strongest full year in the available data at ■555,192.
- 33.6% of transactions have a discount applied, creating a clear area for promotion-performance monitoring.
- Three actions: discount optimization, high-value segment focus, and continuous BI monitoring.

3. Business Objective & Decision Context

- Convert transaction-level data into management-ready evidence for revenue growth, operational focus and recurring KPI monitoring.
- Use a consistent KPI dictionary and data-quality contract before decision-making.
- Combine descriptive analysis, statistical exploration and an interactive dashboard.
- Keep unsupported metrics out of executive decisions: the dataset does not contain profit, CAC, churn or geography fields.

4. Analytical Methodology

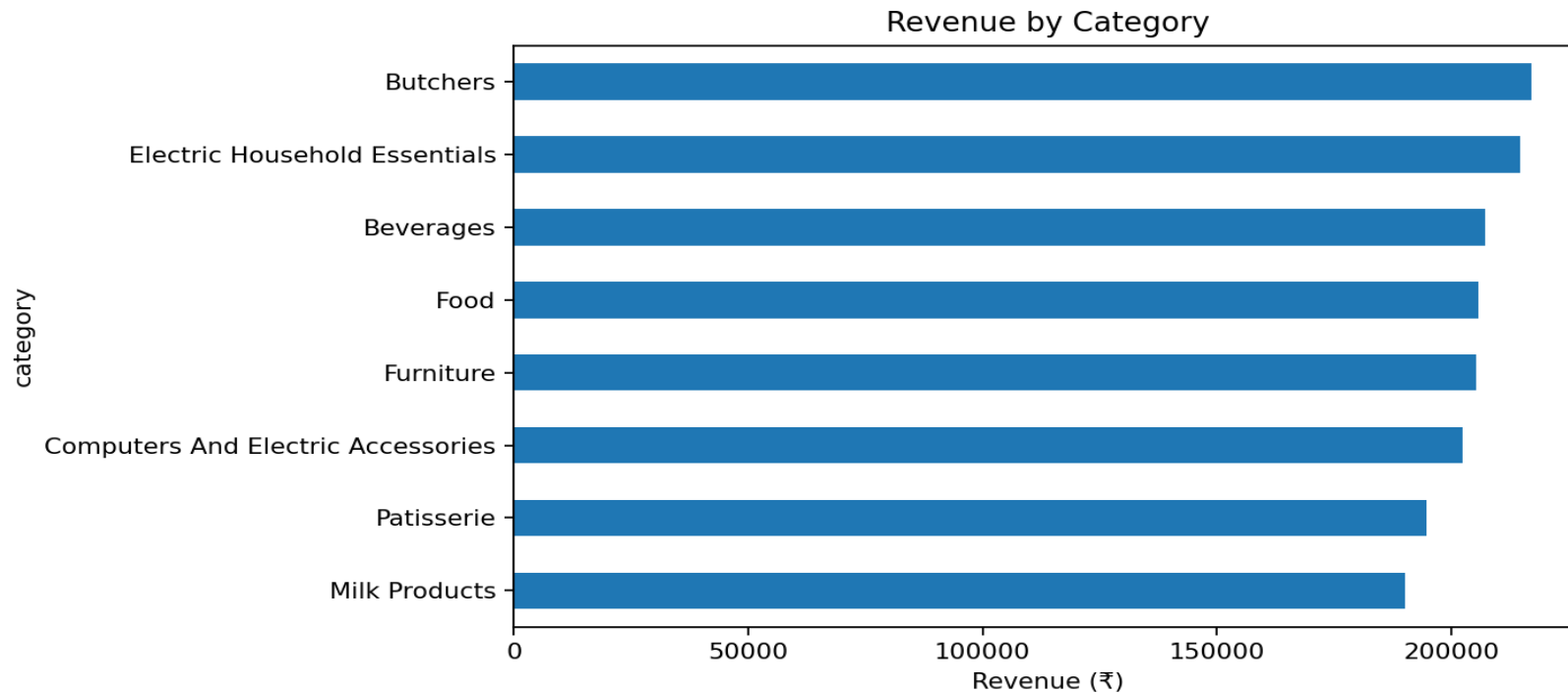
- Task 02: KPI definitions, data-quality profiling and quality thresholds.
- Task 03: missing-value handling, type standardization, date parsing, validation and feature engineering.
- Task 04: descriptive statistics, distributions, correlations, outlier analysis and hypothesis testing framework.
- Task 05: Streamlit + Plotly interactive dashboard with time, category, payment and discount filters.
- Final step: translate observed patterns into executive actions and scenario-based ROI assumptions.

5. Executive KPI Snapshot

Metric	Result
Revenue	■1,636,956
Transactions	12,575
Customers	25
Average Order Value	■130.18
Units Sold	69,828
Discount-applied transactions	33.6%

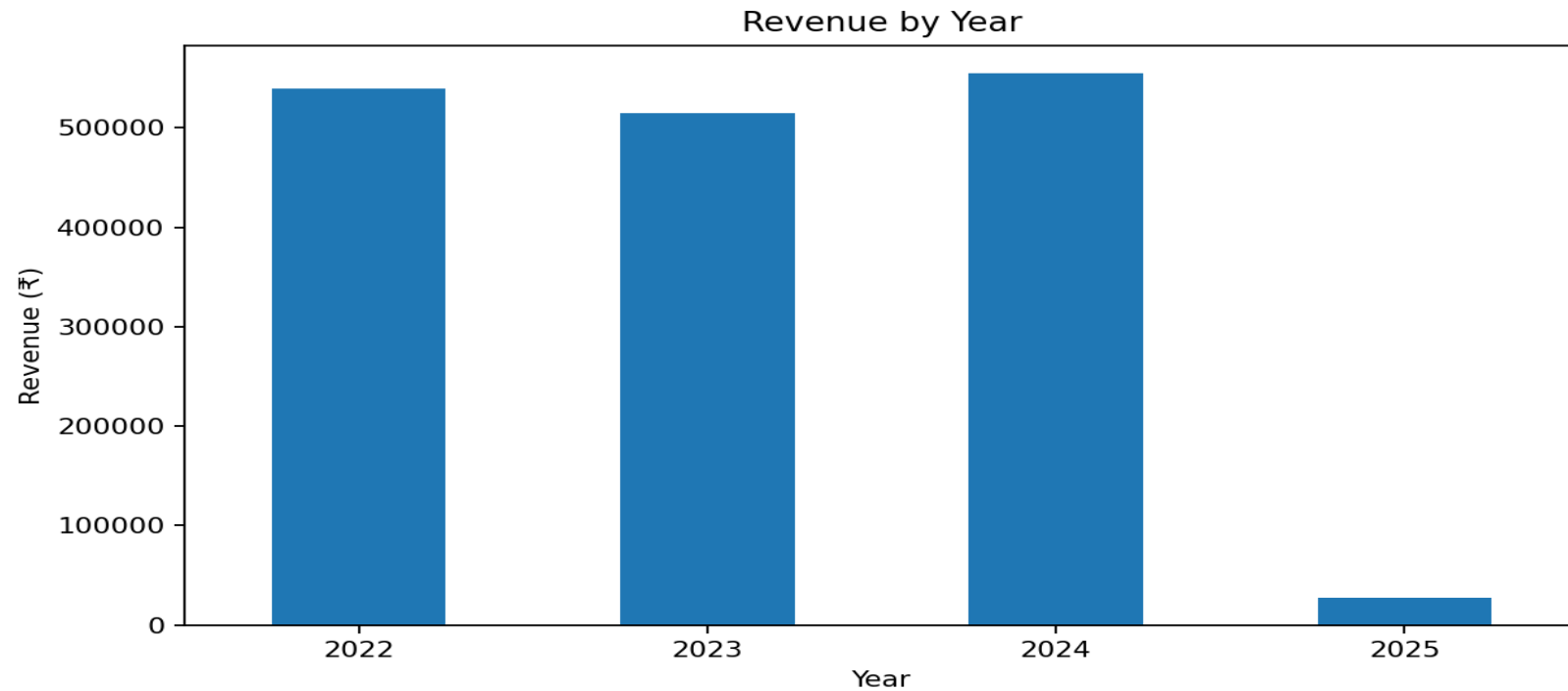
6. Key Revenue Drivers

- Top category: Butchers — ■217,020.
- Second category: Electric Household Essentials — ■214,638.
- Third category: Beverages — ■207,194.
- Cash is the largest payment channel by revenue — ■567,196.
- Management implication: category and channel views should be monitored together when planning promotions and inventory.



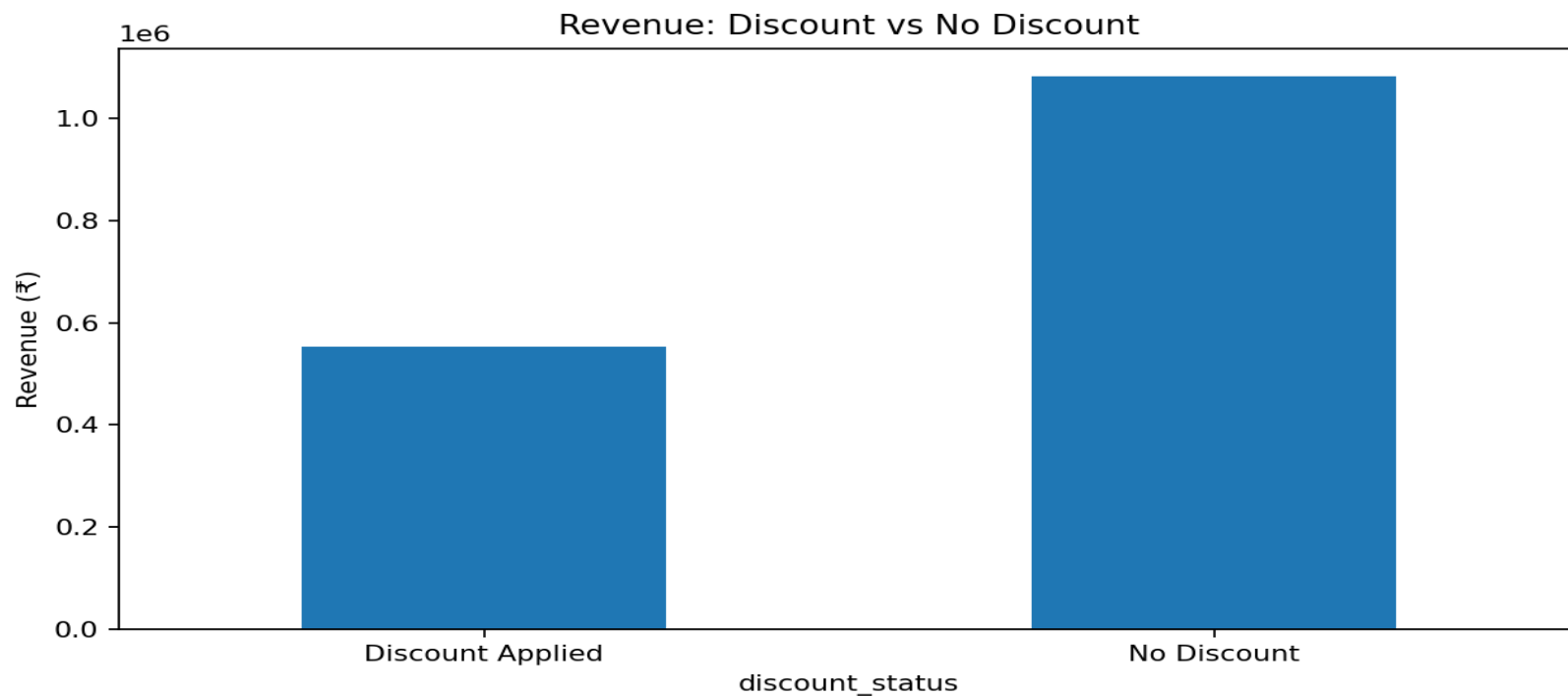
7. Time-Based Performance

- 2024 revenue: ₹555,192; 2022: ₹539,542; 2023: ₹515,206.
- 2025 contains only a small partial set of records in the supplied data, so it should not be compared directly with the full-year periods.
- Monthly variation supports time-based dashboard monitoring rather than relying on a single annual figure.



8. Discount & Promotion Risk Area

- 33.6% of transactions have a discount applied.
- Discount-applied transactions account for ■553,761 revenue; non-discount transactions account for ■1,083,195.
- Because the dataset does not contain profit or discount percentage/value fields, discount profitability cannot be conclusively measured.
- Recommended control: pilot category-level discount thresholds and monitor AOV, revenue and repeat behavior before scaling.



9. Data & Measurement Risks

- Profitability cannot be assessed because no profit/cost field exists in the supplied cleaned dataset.
- CAC and churn are unavailable because acquisition-cost and customer-lifecycle fields are absent.
- Geographic heatmaps cannot be supported because no location field exists.
- 2025 is partial in the supplied data and should not be treated as a comparable full-year benchmark.
- Unknown item labels contribute material revenue; item-level reporting should improve source-data completeness.

10. Three Strategic Recommendations

- 1. DISCOUNT OPTIMIZATION — Establish category-specific promotion rules; test before broad rollout.
- 2. SEGMENT FOCUS — Allocate inventory and promotional attention to consistently high-revenue categories and strong time periods.
- 3. BI GOVERNANCE — Make the dashboard and data-quality checks part of recurring management reporting.

11. ROI Projection & 90-Day Roadmap

Initiative	Scenario assumption	Projected value*	Timing
Discount optimization	5% revenue-equivalent uplift	■81,848	0–3 months
Segment optimization	3% revenue-equivalent uplift	■49,109	3–6 months
BI monitoring	1% value/productivity scenario	■16,370	0–3 months

*Scenario planning only — not measured ROI. Replace assumptions with pilot results before investment approval.

12. Executive Action Plan

- Days 0–30: validate data definitions, confirm category hierarchy, establish baseline KPI dashboard and select promotion pilot.
- Days 31–60: run discount and segment experiments; compare revenue and AOV against baseline.
- Days 61–90: formalize successful controls, automate recurring KPI/data-quality checks and review results with management.
- Decision cadence: monthly executive dashboard review with documented actions, owners and follow-up metrics.
- Success measures: revenue, transactions, AOV, category mix, discount penetration and data-quality compliance.